

Supplementary Figure S14 | Checkpoint, resource, access, and closure governance

(a) Stage ledger

Stage00	CONDITIONAL
Stage01	V2 FAIL
Stage02	STATIC ROUTE CLOSED
Stage03	DYNAMIC ROUTE PAUSED
Stage04	TASK SIGNAL PAUSED
Stage05	OPTIMIZER COVERAGE NOT QUALIFIED
Stage06	FORMAL TRAINING FAIL
Stage07	RETRAINING FAIL
Stage08	COVERAGE POOL NOT QUALIFIED

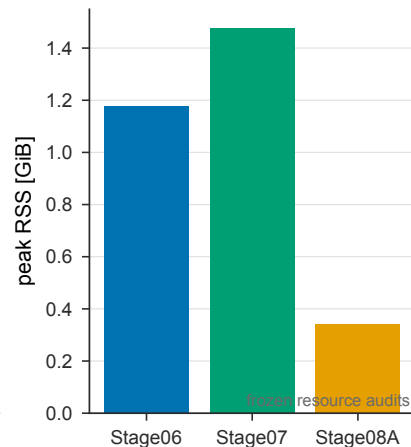
Stage00—Stage08A

(b) Final claim/access flags

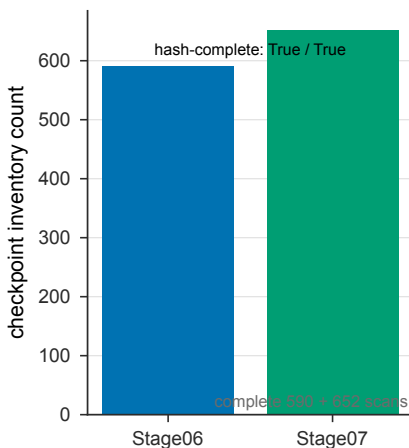
optimizer update qualified	TRUE
autonomous rollout qualified	FALSE
dynamic architecture verified	TRUE
defect target qualified	TRUE
formal training executed	TRUE
trained solver qualified	FALSE
full route closed	TRUE
restricted test evaluated	FALSE
final cycle executed	TRUE
transformer-superiority claim / unsupported	FALSE
V2 baseline restored	FALSE
zero identity verified	TRUE

exact frozen booleans

(c) Resource execution



(d) Checkpoint governance



(e) Final-cycle zero-action ledger

predictions read	0
model forwards	0
optimizer instances	0
optimizer steps	0
parameter updates	0
training runs	0

precomputed coverage only

(f) Immutable closure boundary

